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On the Quality Criteria for Microwave Absorbing Materials

Shengyu Yang, Peng Wu, Yanfei Sheng, Yiwen Dong, Zhiwei Li, Tao Wang, Liang Qiao, and Fashen Li**

The majority of published studies evaluating microwave absorbing (MA) materials focus on calculating reflectivity, identifying materials with lower reflectivity as having "good absorption capability". However, Rozanov et al. posited that reflection minimum depth is insufficient as the quality criteria—a viewpoint that remains under-recognized in scientific discourse so far. This study leverages the quarter-wavelength wavelength relationship in conjunction with transmission line theory to engineer various microwave absorbers composed of ferromagnetic or dielectric materials. The results indicate that each of these absorbers can achieves perfect matching ($Z_{in}/Z_0 = 1$), suggesting that, in practice, they can all achieve the best reflection loss (RL) value of negative infinity, thereby affirming Rozanov ′s perspective. In order to be able to better evaluate MA materials, it is identified that the squared refractive index at perfect matching, denoted as $(\mu\epsilon)_p$, serves as a robust quality metric. Experimental results indicate that materials with a higher $(\mu\epsilon)_p$ value exhibit superior absorption performance. Furthermore, this metric offers valuable insights for the design of low-frequency thin-layer absorbers and for identifying better MA materials. These findings underscore an indissociable link between MA device performance and MA material characteristics.

1. Introduction

In recent years, microwave absorbing (MA) materials have received growing attention owing to their broad applicability in electromagnetic interference shielding, wireless communications, and the protection of electronic devices.[1–4] These materials play a crucial role in addressing electromagnetic pollution, an escalating concern particularly with the rapid deployment of advanced technologies such as 5G, which—despite offering unprecedented convenience—has introduced new risks to human health and environmental safety.[5–8] Consequently, the pursuit of high-performance MA materials with integrated efficiency, stability, and adaptability has become increasingly imperative.[9–12] The current array of MA materials in the literature is diverse, and the notion that "everything can be wave-absorbing" is not an exaggeration. From the prominent attention paid to graphene,[13] carbon fibers,[14] and carbon nanotubes[15] in the past few years, to various oxides,[16–18] sulfides,[19] phosphides,[20,21] polymers,[22] metals/alloys,[23–26] and now focusing on aerogels,[27,28] nanomaterials,[29] and MOFs[30,31] and biomass[32] materials, all can serve as "a promising candidate of MA materials". For instance, carbon-based materials such as graphene, carbon nanotubes, and MXenes are known for their high conductivity and multiple relaxation mechanisms, enabling strong dielectric loss. However, their impedance mismatching often limits their absorption at lower frequencies.[33] Biomass-derived porous carbons have emerged as sustainable alternatives, offering hierarchical structures and adjustable permittivity, yet still face challenges in achieving broadband absorption under thin-layer configurations.[34,35] In fact, over 90% of the absorption frequency of MA materials derived from the aforementioned substances fall within the X—band or even higher frequency. This underscores a long-standing and critical challenge: designing MA materials capable of strong absorption in the low-frequency regime while retaining ultrathin and lightweight characteristics.

Furthermore, a widely held belief among researchers is that lower reflection loss (RL) values indicate superior microwave absorption performance, and RL is often treated as a definitive metric for evaluating MA materials. However, this perspective is flawed. As early as five years ago, Rozanov's team and Dr. Chen's team were the first to raise questions about RL in microwave absorbers. Rozanov[36] contended that the superior radar absorption performance is not just about obtaining low RL values. This issue fundamentally involves the appropriate matching of the material's absorptive parameters. One cannot simply evaluate MA materials by concluding that the RL value of −30 dB outperforms −20 dB in terms of microwave absorption performance, nor can it be straightforwardly asserted that −50 dB is better than −40 dB. Dr. Chen[37] further argued that improving the precision of the frequency and thickness parameters around a minimal RL

S. Yang, P. Wu, Y. Sheng, Y. Dong, Z. Li, T. Wang, L. Qiao, F. Li
Institute of Applied Magnetism
Key Laboratory for Magnetism and Magnetic Materials of Ministry of Education
Lanzhou University
Lanzhou 730000, P. R. China
E-mail: qiaoliang@lzu.edu.cn; lifs@lzu.edu.cn

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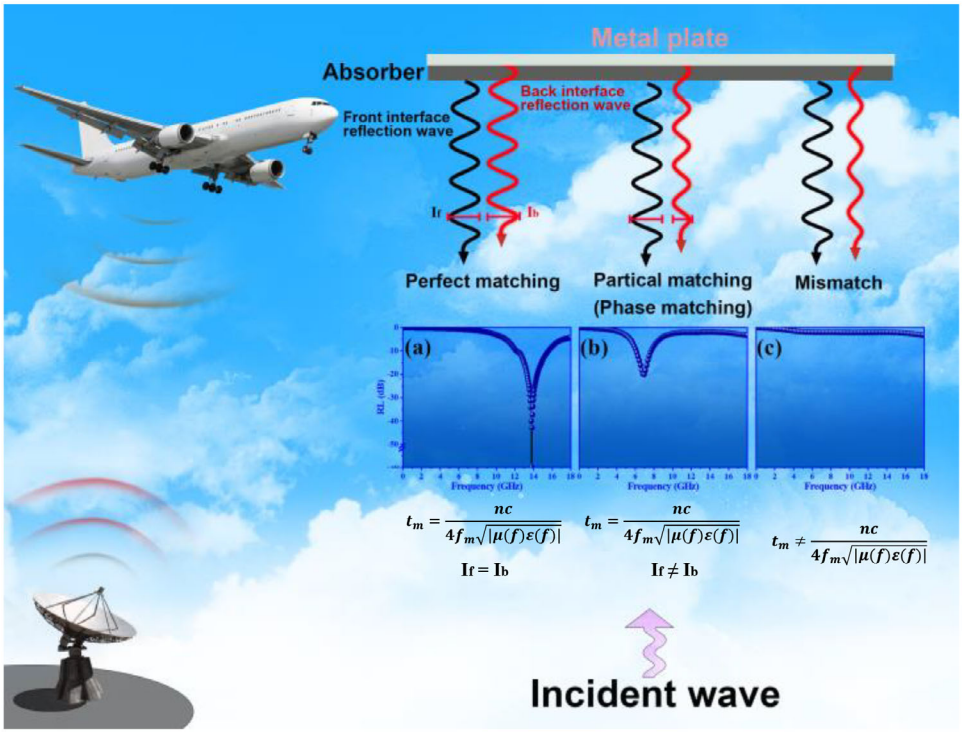


Figure 1. The interaction of reflected waves at the front interfaces and back interfaces leads to three matching scenarios and corresponding RL curves.

point can decrease the resulting RL value. This suggests the minimal RL value is potentially a calculation artifact rather than an intrinsic material property. The issues identified by Rozanov and Dr. Chen are both pivotal and scientifically robust. Regrettably, these critical insights have been largely neglected in the past five to six years. In other words, the majority of researchers continue to lack a well-defined framework for determining what qualifies as a better MA material. This has perpetuated the misconception that "everything can be microwave-absorbing". This has led to a misallocation of research resources and hindered meaningful progress in addressing one of the field's most pressing challenges: achieving strong low-frequency absorption with ultrathin materials. To move forward, it is imperative to establish a robust quality evaluation standard that directly addresses the key performance requirements of next-generation MA materials.

In this study, a comprehensive analysis and in-depth discussion were conducted on the material matching problem mentioned by Rozanov, which is intricately linked to the design methodologies of absorbers. Ferromagnetic materials (Fe_4N, Fe_3Si, and nickel-zinc ferrite, etc.) and dielectric materials (CeO_2,) were chosen as subjects of investigation. The results indicate that with a well-designed absorber, perfect matching can be achieved, resulting in RL approaching negative infinity for each material. It is vital to emphasize that RL is a parameter of the absorbing device and should not be used to evaluate the intrinsic quality of a material. This is akin to the inappropriate use of inductance values as quality criteria of soft magnetic materials. Accordingly, an urgent need exists to formulate robust standards for the accurate and effective evaluation of material quality. To address the most critical challenges in contemporary absorber research, we propose adopting the squared refractive index as a reliable and effective quality criterion. Whether tackling the formidable challenge of low-frequency absorption or satisfying the stringent thin-layer requirements in advanced flight weaponry, the squared refractive index provides a new idea for absorber design.

2. Results and Discussion

2.1. Interface Reflection Models and Impedance Matching Relationship

Upon incidence of electromagnetic (EM) waves on the absorber, which is mounted on a perfect conductive metal backing plate, two distinct interfacial reflection waves are observed. The first occurs at the interface between air and the absorber, while the second takes place at the interface between the absorber and the metal backing plate. Characterized by high electrical conductivity, the metal backing plate efficiently reflects all incident electromagnetic waves (as shown in **Figure 1**). Consequently, the reflection loss attributable to the absorber is quantifiable and can be elucidated as follows:[38,39]

$$\mathrm{RL\,(dB)} = 20log_{10}\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right| \tag{1}$$

$$Z_c = \frac{Z_{in}}{Z_0} = \sqrt{\frac{\mu\,(f)}{\varepsilon\,(f)}}\tanh\left[\left(i\frac{2\pi ft}{\mathrm{c}}\right)\sqrt{\mu\,(f)\,\varepsilon\,(f)}\right] \tag{2}$$

where Z_{in}, Z_0, μ, ϵ, i, f, t, and c are the input impedance at the absorber surface, the impedance of the air, the complex

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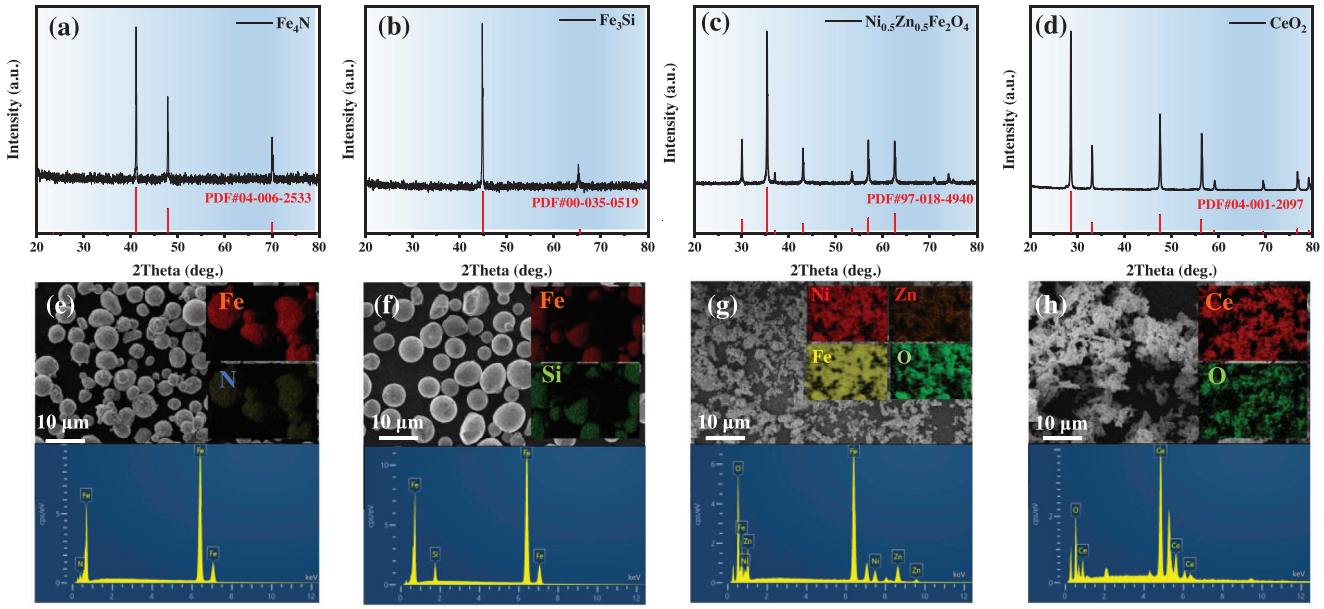


Figure 2. XRD a–d), microstructural morphology, and energy-dispersive spectroscopy spectra e–h) patterns of Fe_4N, Fe_3Si, Ni—Zn ferrite, and CeO_2.

permeability and permittivity, the unit complex number, the frequency of the microwave, the thickness of the sample, and the velocity of the light, respectively. As can be easily seen, the complex permeability and complex permittivity directly determine the complex impedance (Z_c) as well as the numerical magnitude of the RL. To obtain an exceptionally low RL value, it is imperative to ensure electromagnetic impedance matching and to achieve a frequency and thickness that satisfies the quarter-wavelength wavelength relationship. The relationship is as follows:[40]

$$t_m = \frac{nc}{4f_m\sqrt{|\mu(f)\,\varepsilon(f)|}}, (n = 1, 3, 5\ldots) \quad (3)$$

where f_m is the frequency at peak RL, t_m is the sample thickness, and n = 1, 3, 5... The quarter-wavelength theory is often utilized to elucidate the shift of RL peaks toward lower frequency with increasing sample thickness. Its more significant utility lies in accurately determining the absorber thickness t_m that achieves phase matching at any given frequency.

As illustrated in Figure 1, the concept of matching encompasses three distinct categories: perfect matching, partial matching (phase matching), and mismatch. When the reflected waves at the front and back interfaces differ by 180° in phase and are unequal in amplitude, the two reflected waves achieve only phase matching. Under these conditions, a relatively modest peak appears on the RL curve, as depicted in Figure 1b. When both the phase matching and equal amplitude conditions ($I_f = I_b$) are simultaneously met, the theoretical RL peak can reach negative infinity, signifying perfect matching. In other words, perfect matching represents a specific case of phase matching. Conversely, mismatch implies that the phase-matching condition is not fulfilled.

Consequently, the critical question arises: how can perfect matching between two reflected waves be achieved? The phase and amplitude of the two reflected waves can be modulated through the transmittance, refractive index, and propagation loss within the absorber. Therefore, this question shifts to designing absorbers that can achieve perfect matching between the two reflected waves. By combining Equations (2) and (3), the relationship between impedance and frequency at phase-matching conditions can be derived. Furthermore, when the impedance $Z_c = 1$, the reflected waves at the front and back interfaces exhibit identical amplitudes, achieving perfect matching at the corresponding frequency. Substituting the matching frequency into Equation (3) determines the matching thickness, thus completing the design of an absorber capable of achieving perfect matching between the two reflected waves.

2.2. Characterization of Structure and Microwave Absorption Performance

X-ray diffraction analysis of the sample's crystalline composition (**Figure** 2a–d) revealed that both the Fe_4N obtained via gas-phase nitridation and the commercial Fe_3Si, Ni—Zn ferrite, and CeO_2 materials exhibited diffraction peaks in precise correspondence with their respective standard PDF reference patterns, confirming the high crystallinity and phase purity of all materials. The microstructural morphology and energy-dispersive spectroscopy (EDS) spectra are presented in Figure 2e–h. The synthesized Fe_4N particles exhibit a distinct, 3D elliptical morphology with an outer layer displaying a cracked eggshell-like texture. This phenomenon arises because nitrogen atom incorporation into the lattice during nitridation induces an increase in unit cell volume, thereby resulting in particle expansion. Furthermore, Fe_3Si particles produced via the atomization method display a comparatively smooth ellipsoidal morphology. Both particle types exhibited a size distribution ranging from 5 to 15 μm. In contrast, Ni—Zn ferrite and CeO_2 particles adopted an irregular, blocky morphology, with particle sizes below 5 μm. The EDS spectra beneath the

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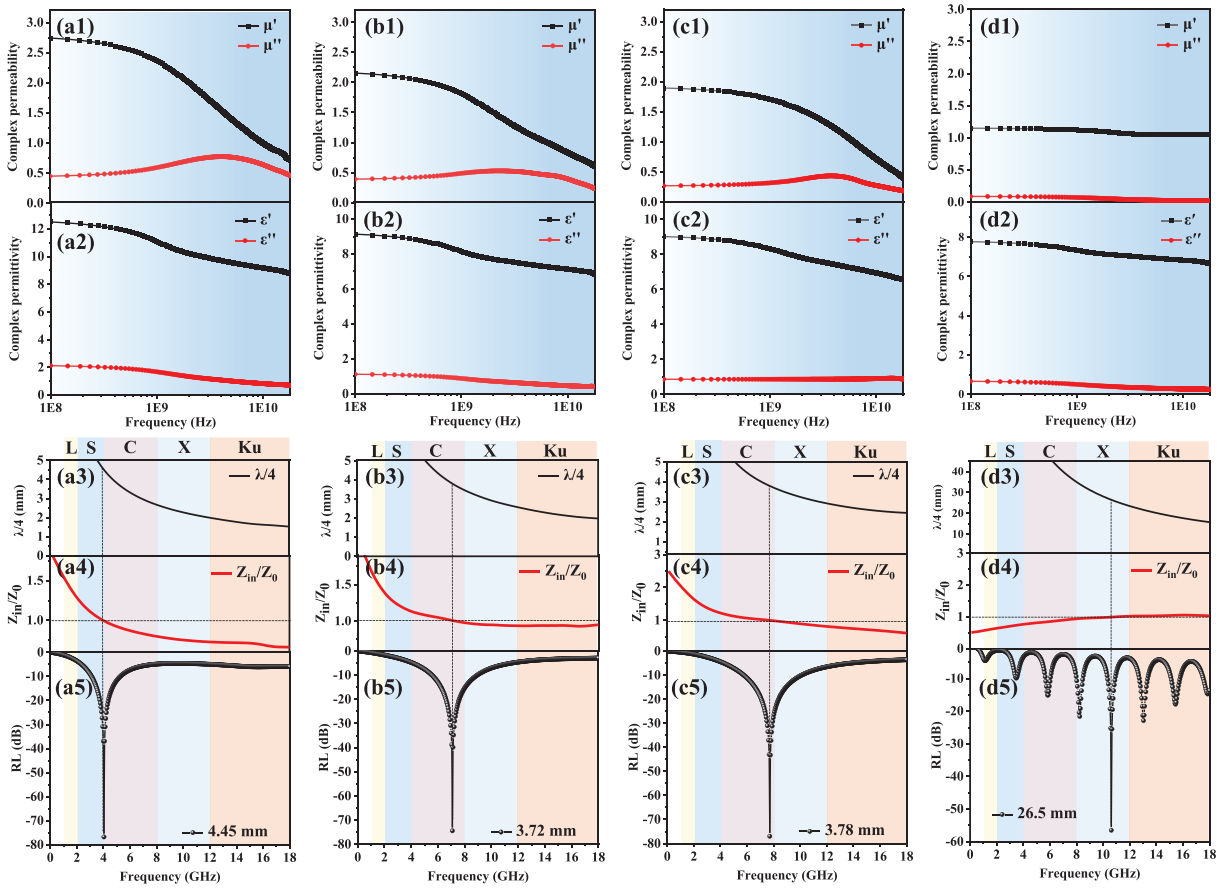


Figure 3. Complex permeability a1, b1, c1, d1), complex permittivity a2, b2, c2, d2), phase matching thickness over frequency (t_m-f_m) curve a3, b3, c3, d3), phase matching impedance over frequency (Z_{in}/Z_0-f_m) curve a4, b4, c4, d4) and reflection loss curve at the perfect matching point a5, b5, c5, d5) of Fe_4N, Fe_3Si, Ni—Zn ferrite and CeO_2.

primary figure further illustrate the elemental characteristics and relative compositions of each sample. Furthermore, X-ray photoelectron spectroscopy (XPS) depth-profiling analyses were conducted on all four materials, as detailed in the supplementary materials. Notably, for the synthesized Fe_4N sample, the Fe $2p_{3/2}$ and Fe $2p_{1/2}$ spectral features were deconvoluted into two distinct doublets. Peaks located at 710.1 eV (Fe $2p_{3/2}$) and 726.6 eV (Fe $2p_{1/2}$) are assigned to Fe^{3+} species, while those at 707.6 eV (Fe $2p_{3/2}$) and 721.3 eV (Fe $2p_{1/2}$) correspond to Fe—N coordinated species, confirming the successful synthesis of γ'-Fe_4N.[41] These XPS findings are consistent with the phase identification results obtained from XRD and EDS analyses. Similarly, the XPS results of other commercial materials show strong consistency with their respective XRD and EDS data.

To further measure electromagnetic parameters, each of the four materials was fabricated into composite samples with a 20% volume fraction. The permeability spectra shown in **Figure** 3a1–d1 indicate substantial differences in the magnetic properties across the four materials. Based on Equation (3), the phase-matching thickness as a function of frequency was plotted, as illustrated in Figure 3a3–d3. This curve demonstrates that at any given frequency, phase matching of the reflected waves at both the front and back interfaces is achieved when the absorber's thickness attains the required matching value. For perfect matching, Equations (2) and (3) are combined, and the impedance versus frequency curve is depicted in Figure 3a4–d4. When the impedance Z_c reaches 1, perfect matching is achieved at the matching frequency and matching thickness.

Using Fe_4N as an illustrative example, Figure 3a4 demonstrates that this material achieves perfect matching at 4 GHz, with a corresponding matching thickness of 4.45 mm as shown in Figure 3a3. By substituting this specific matching frequency and thickness into Equation (1) to calculate RL, Figure 3a5 reveals a RL value for Fe_4N approaching −80 dB. If RL value is used as the evaluation criterion, Fe_4N's performance in microwave absorption could be regarded as outstanding. In fact, it is not. As long as perfect matching is achieved, virtually any material can exhibit extremely low RL values. Following the same methodology, the perfect matching conditions for Fe_3Si, Ni—Zn ferrite, and CeO_2 were identified, with the resulting RL curves presented in Figure 3a5,b5,c5,d5. It is observed that all materials exhibit RL values exceeding −50 dB, a result that is not coincidental. The issue raised by Rozanov, therefore, demands serious consideration. It is important to note that achieving an RL value approaching negative infinity remains a theoretical limit, difficult to attain experimentally. Because the precision of the

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permeability and permittivity measurements, coupled with the exactitude of frequency and thickness inputs, renders the electromagnetic impedance matching exceedingly close to, yet not exactly, 1. This explains Professor Chen's assertion that the seemingly minimal RL values observed might be computational artifacts. When the RL value exceeds −40 dB, indicating an absorption rate above 99.99%, and the impedance deviation is constrained within 2% ($0.98<|Z_{in}/Z_0|<1.02$), the absorber can be regarded as having achieved perfect matching.

The reflection loss curve of the dielectric material CeO_2 exhibits intriguing characteristics. Unlike ferromagnetic materials, CeO_2 displays multiple absorption peaks with varying intensities, as illustrated in Figure 3d5. Notably, perfect matching is achieved at 10.5 GHz with an absorber thickness of 26.5 mm, producing an RL value of −56.5 dB. The remaining peaks are attributed to phase matching occurring at various frequency. It is an undeniable fact that dielectric materials can also achieve remarkably low RL value. However, despite its low RL value, the matching thickness required for CeO_2 is 26.5 mm, an order of magnitude higher than that required for the other three ferromagnetic materials. This phenomenon reaffirms, once again, that any material is capable of achieving remarkably low RL values, rendering RL value insufficient as a criterion for assessing absorber performance. Furthermore, it underscores the inherent complexity in assessing absorber devices, for which no universally accepted solution currently exists. This study seeks to provide an innovative and effective resolution to this limitation.

2.3. Evaluation of Absorber Performance and Quality Criteria of MA Materials

Addressing the contemporary challenge of achieving better absorption in low-frequency thin layers necessitates an in-depth exploration of perfect matching conditions. Leveraging the quarter-wavelength relationship and transmission line theory, the perfect matching frequency (f_m) and thickness (t_m) can be obtained. The absorption thickness factor $(f_m t_m)_p$, defined as the product of the perfect matching frequency and thickness. The derived expression reveals that, the smaller absorption thickness factor under the same matching frequency corresponds to a thinner absorber design—an aspect critical to aerospace applications. In addition, the smaller absorption thickness factor under the same matching thickness indicates that the absorber can work in the lower frequency region, which provides a new idea for solving the current problem of the difficulty of low-frequency microwave absorption. Therefore, a lower absorption thickness factor $(f_m t_m)_p$ largely indicates that the absorber possesses superior microwave absorption performance.

Further, a direct correlation between the absorption thickness factor $(f_m t_m)_p$ and the material's intrinsic parameters can be derived, as illustrated below:

$$(\mu\varepsilon)_p = \left(\frac{nc}{4(f_m t_m)_p}\right)^2, (n = 1, 3, 5, \cdots) \quad (4)$$

Here, f_m and t_m are the perfect matching frequency and thickness, respectively. $(\mu\epsilon)_p$ is the squared refractive index at perfect matching. The absorption thickness factor, defined as the product of f_m and t_m, essentially serves as a device parameter of the absorber, whereas the squared refractive index $(\mu\epsilon)_p$ fundamentally encapsulates the intrinsic material properties. It can be said that the larger the $(\mu\epsilon)_p$ of the material is the better the microwave absorption performance of the prepared microwave absorber, and the larger the $(\mu\epsilon)_p$ is, the more conducive to solving the current problem of strong absorption of low-frequency thin layers. This assists researchers in designing more Figure 1 efficient microwave absorbers from the characteristics of the materials.

Building on the previous analysis, it is essential to further verify the appropriateness of utilizing the square refractive index as a quality standard. The 3D reflection loss curves for ferromagnetic materials, including Fe_4N, $FeSi_{3.5}$, Ni—Zn ferrite, as well as Fe, FeCo, FeSiCr, are presented in **Figure** 4a1–f1 to facilitate a clearer examination of the microwave absorption characteristics of each material. Figure 4a2–f2 display the corresponding 2D projection diagrams. Fe, FeCo, and FeSiCr were added for subsequent comparisons, and their electromagnetic parameters are shown in the Supporting Information. First, a visual assessment reveals that each material achieves notably low RL values. In the 3D plots, the frequency and thickness corresponding to the minimum RL point represent the conditions for achieving perfect matching in the absorber. Second, the projection diagrams delineate the effective bandwidth at −10 dB and the ultra-strong absorption bandwidth at −40 dB using dashed lines, indicating that the optimal microwave absorption frequency of each material occurs solely within these defined ranges. Indiscriminate application of these materials would likely inhibit their peak absorption performance.

As illustrated in Figure 4g, Material B's perfect matching frequency resides within the Ku band, suggesting that a RL approaching negative infinity is achievable at a specific frequency point in this range. Nevertheless, this also implies that it may exhibit suboptimal absorption performance in the S band, thus limiting its direct comparison with Material A, which possesses a perfect matching frequency within the S band, despite both materials having identical squared refractive index. Therefore, it would be beneficial to systematically categorize each material based on its perfect matching frequency into distinct frequency band databases for subsequent quality evaluation. For example, the perfect matching frequency of Fe_4N, $FeSi_{3.5}$, Ni—Zn ferrite, and CeO_2 are 4 GHz, 7.26 GHz, 7.71 GHz, and 10.93 GHz, respectively, categorizing them into the C—band, C—band, C—band, and X—band as their optimal operational bands. **Table** 1 summarizes microwave absorption parameters for each material, including their corresponding optimal operating frequency bands. To further investigate the relationship between the squared refractive index and absorption performance, we initially focus on the Fe, Fe_4N, Fe_3Si, and Ni—Zn ferrite materials in the C—band (4–8 GHz), selecting a central frequency of 6 GHz as the observation frequency. This is because, at other frequency bands, these materials may not exhibit optimal absorption properties. At 6 GHz, the squared refractive index for these four materials are 32.37, 16.89, 7.61, and 6.6, respectively. As depicted in Figure 4f, Fe exhibits a higher square of the refractive index, enabling it to achieve a comparable effective bandwidth and peak absorption strength to the other materials, despite its relatively small thickness of only 2.15 mm. In contrast, the Ni—Zn ferrite, with a lower square of the refractive index,

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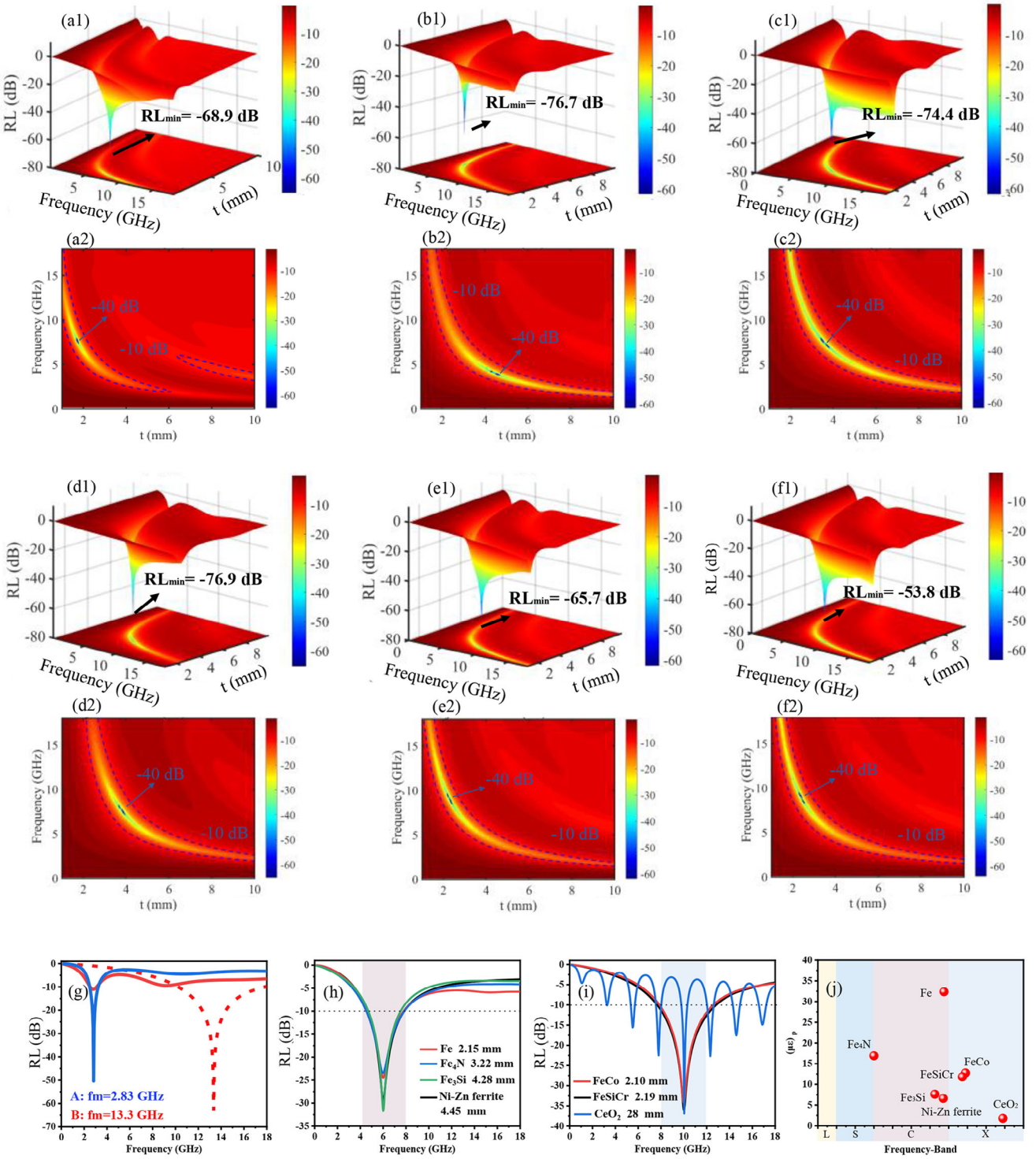


Figure 4. 3D and 2D representation of RL values of Fe (a1,a2), Fe_4N (b1,b2), Fe_3Si (c1,c2), Ni—Zn ferrite (d1,d2), FeCo (e1,e2), FeSiCr (f1,f2). Reflection loss curves g) of material B at the perfect matching frequency of material A. Reflection loss curves of Fe, Fe_4N, Fe_3Si, Ni—Zn ferrite at 6 GHz h) and FeCo, FeSiCr, CeO_2 at 10 GHz i). Perfect Match Frequency-Band Database of Microwave Absorbing Materials j).

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Table 1. Comparison of microwave absorption properties of Fe, Fe_4N, Fe_3Si, Ni—Zn ferrite, FeCo, FeSiCr, CeO_2.

	f_m	t_m	$(f_m t_m)_p$	$(\mu\epsilon)_p$	Wave Band	At the Center Frequency of the Corresponding Band.		
						t/mm	Δf/GHz	RL/dB
Fe	7.75	1.70	13.18	32.37	C	2.15	3.18	−24.5
Fe_4N	4	4.56	18.24	16.89	C	3.22	3.40	−23.3
Fe_3Si	7.26	3.73	27.08	7.61	C	4.28	2.91	−32.2
Ni-Zn ferrite	7.71	3.79	29.22	6.60	C	4.45	3.17	−29.7
FeCo	8.9	2.36	21.00	12.75	X	2.10	4.66	−35.3
FeSiCr	8.73	2.5	21.83	11.80	X	2.19	5.23	−36.0
CeO_2	10.9	25.7	281.2	1.78	X	28.2	1.68	−36.9

necessitates a greater thickness (4.45 mm) to attain comparable absorption performance. The same method applies effectively in the X—band as well.

This demonstrates that materials with higher square refractive indices exhibit significant advantages in absorption performance, particularly in terms of reducing material thickness and designing compact microwave-absorbing devices. Ultimately, by mapping all the materials discussed in this study onto Figure 4j, the microwave absorption performance of these materials is clearly illustrated.

3. Conclusion

In conclusion, experiments were conducted to microwave absorption performance and related electromagnetic parameters of representative materials: ferromagnetic, antiferromagnetic, and pure dielectric materials. The results are as follows:

1) The performance and applicability of microwave-absorbing devices cannot be evaluated by the magnitude of reflection loss, because any magnetic or dielectric material can theoretically and practically achieve perfect matching, resulting in the RL value of negative infinity. Consequently, new quality criteria are required in the domain of MA materials.
2) When pure dielectric materials are used as microwave absorbers, their absorption mechanism also follows the interface reflection model. Specifically, the phase-matching condition for the formation of absorption peaks is $t_m = \frac{n\lambda}{4f_m\sqrt{|\mu(f)\epsilon(f)|}}$, where for dielectric materials, the formula has $\mu = 1$.
3) In this work, the squared refractive index at perfect matching $(\mu\epsilon)_p$proposed serves as a more robust quality metric for evaluating the microwave absorption properties of materials. A comparative analysis with existing materials reveals that materials with larger squared refractive index $(\mu\epsilon)_p$exhibit superior microwave absorption performance. Specifically, at a given matching frequency, a smaller absorption thickness factor $(f_m t_m)_p$ corresponds to a thinner absorber design. At an equivalent matching thickness, a smaller absorption thickness factor $(f_m t_m)_p$ suggests that the absorber is capable of operating in the lower frequency range, presenting a novel solution to the ongoing challenge of low-frequency microwave absorption. Building on this new standard, the study offers a critical theoretical framework and practical insights for the design and optimization of microwave absorbing materials.

4. Experimental Section

Fe_4N particles were synthesized via gas-phase nitridation by exposing high-purity iron powder to a mixed atmosphere of hydrogen and ammonia at 600 °C for 5 h. Fe_3Si (CHANGSHA TIJO METAL MATERIAL CO., LTD), Ni—Zn ferrite (Hebei Chaoyue Metal Products Co., Ltd.), and CeO_2 (Macklin) were all procured from commercial suppliers. The crystal structure and morphology analyses of Fe_4N, Fe_3Si, Ni—Zn ferrite, and CeO_2 were performed by X-ray diffraction (XRD), scanning electron microscopy (SEM, Apollo 300). In order to measure the complex permeability and complex permittivity from 0.1 to 18 GHz using a vector network analyzer (VNA, Agilent N5247A), the particles were compressed with polyurethane under a pressure of 4 MPa to form ring-shaped samples with an inner diameter of 3.04 mm and an outer diameter of 7.00 mm. All composite materials used a volume fraction of 20% (See Supporting Information S1 for detailed preparation procedures.).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data availability statement

The data that support the findings of this study are available from thecorresponding author upon reasonable request.

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Keywords

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